**TASK 1: DNA/Protein Sequence Analysis Using BLAST**

## Title

BLAST Analysis of Human Hemoglobin Beta Protein (HBB)

## Objective

To identify homologous protein sequences of Human Hemoglobin Beta Protein using BLAST and analyze sequence similarity based on alignment score, query coverage, identity percentage, and E-value.

## Introduction

BLAST (Basic Local Alignment Search Tool) is a bioinformatics tool used to compare biological sequences and identify similar sequences in databases. In this experiment, the Human Hemoglobin Beta Protein sequence was obtained from UniProt and analyzed using NCBI BLASTP to identify homologous proteins.

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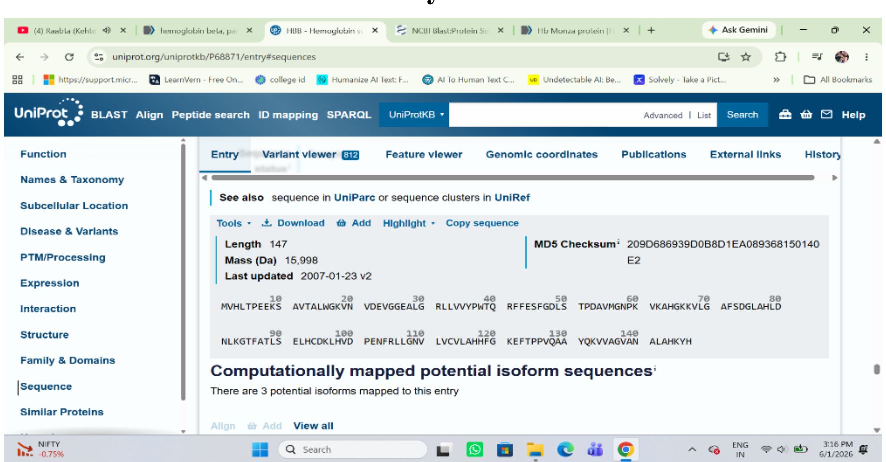
## Protein Information

| Parameter | Details |
|---|---|
| Protein Name | Hemoglobin Subunit Beta |
| Gene Name | HBB |
| Organism | Homo sapiens |
| UniProt ID | P68871 |
| Sequence Length | 147 Amino Acids |

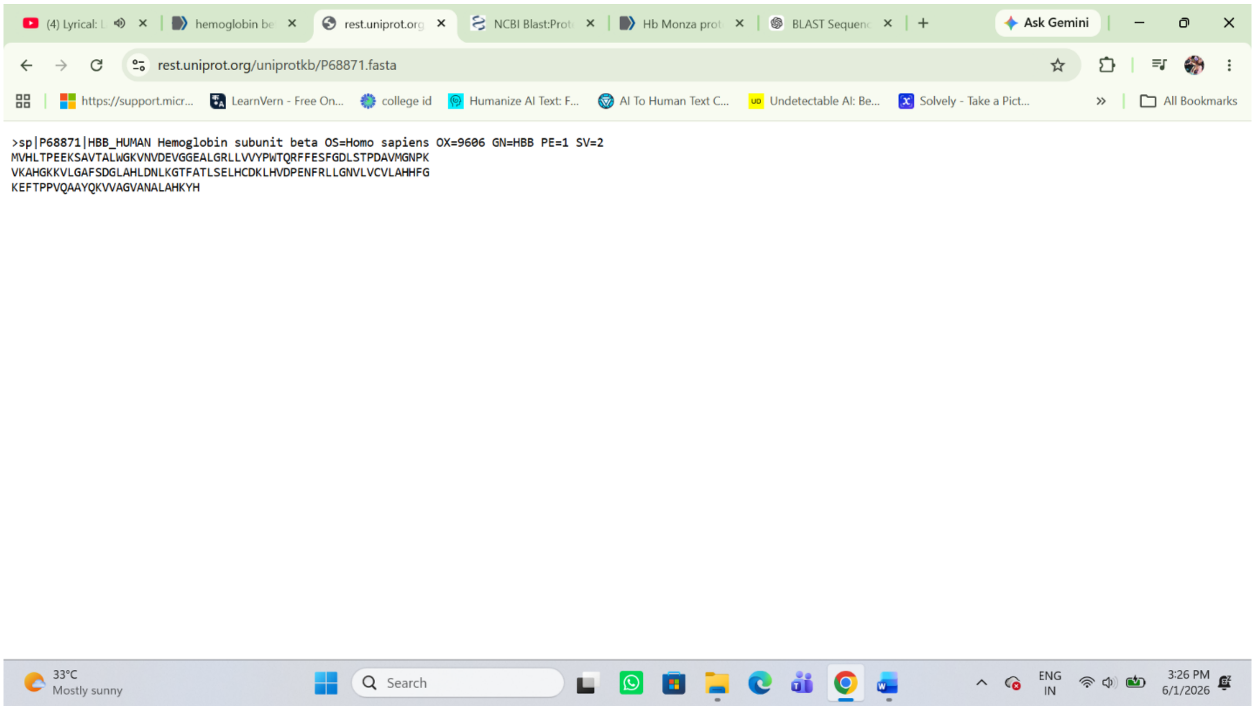
## Screenshot 1: UniProt Protein Entry



## FASTA Sequence

The protein sequence was downloaded in FASTA format from UniProt and used as the query sequence for BLAST analysis.

## Screenshot 2: FASTA Sequence

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## Methodology

1. Human Hemoglobin Beta Protein sequence was downloaded from UniProt.
2. The FASTA sequence was copied.
3. NCBI BLASTP was opened.
4. The FASTA sequence was pasted into the query box.
5. Default BLAST parameters were used.
6. BLAST search was executed.
7. Homologous protein sequences were analyzed.
8. Query coverage, identity percentage, E-value, and alignment score were recorded.

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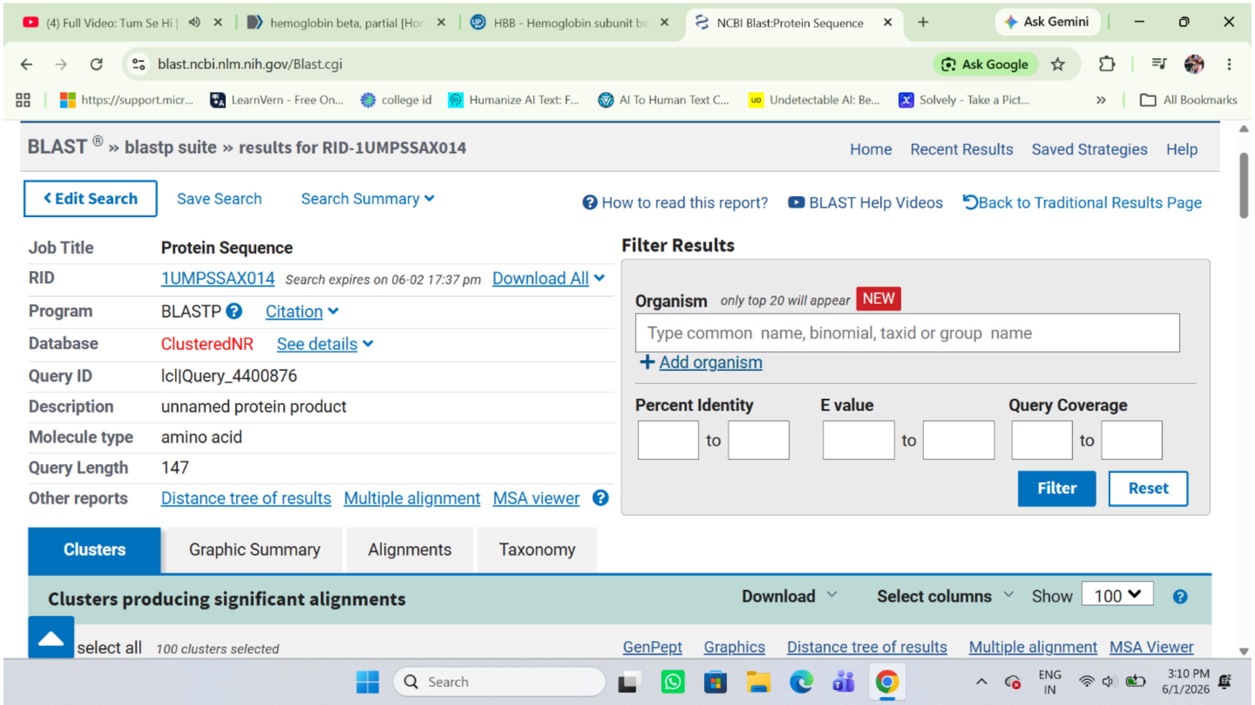
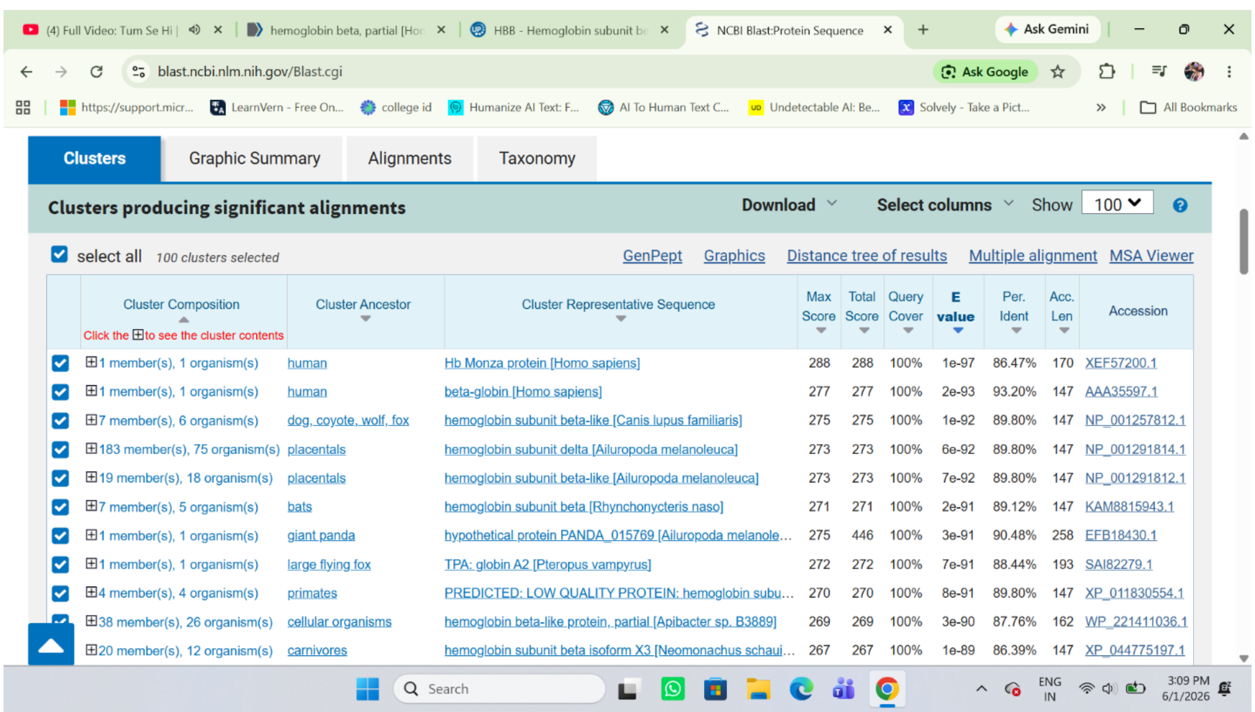
## BLAST Results

## Screenshot 3: BLAST Results Page (Clusters)







Clusters producing significant alignments

select all 100 clusters selected

| Cluster Composition | Cluster Ancestor | Cluster Representative Sequence | Max Score | Total Score | Query Cover | E value | Per. Ident | Acc. Len | Accession |
|---|---|---|---|---|---|---|---|---|---|
| 1 member(s), 1 organism(s) | human | Hb Monza protein [Homo sapiens] | 288 | 288 | 100% | 1e-97 | 86.47% | 170 | XEF57200.1 |
| 1 member(s), 1 organism(s) | human | beta-globin [Homo sapiens] | 277 | 277 | 100% | 2e-93 | 93.20% | 147 | AAA35597.1 |
| 7 member(s), 6 organism(s) | dog, coyote, wolf, fox | hemoglobin subunit beta-like [Canis lupus familiaris] | 275 | 275 | 100% | 1e-92 | 89.80% | 147 | NP_001257812.1 |
| 183 member(s), 75 organism(s) | placentals | hemoglobin subunit delta [Ailuropoda melanoleuca] | 273 | 273 | 100% | 6e-92 | 89.80% | 147 | NP_001291814.1 |
| 19 member(s), 18 organism(s) | placentals | hemoglobin subunit beta-like [Ailuropoda melanoleuca] | 273 | 273 | 100% | 7e-92 | 89.80% | 147 | NP_001291812.1 |
| 7 member(s), 5 organism(s) | bats | hemoglobin subunit beta [Rhynchonycteris naso] | 271 | 271 | 100% | 2e-91 | 89.12% | 147 | KAM8815943.1 |
| 1 member(s), 1 organism(s) | giant panda | hypothetical protein PANDA_015769 [Ailuropoda melanole… | 275 | 446 | 100% | 3e-91 | 90.48% | 258 | EFB18430.1 |
| 1 member(s), 1 organism(s) | large flying fox | TPA: globin A2 [Pteropus vampyrus] | 272 | 272 | 100% | 7e-91 | 88.44% | 193 | SAI82279.1 |
| 4 member(s), 4 organism(s) | primates | PREDICTED: LOW QUALITY PROTEIN: hemoglobin subu… | 270 | 270 | 100% | 8e-91 | 89.80% | 147 | XP_011830554.1 |
| 38 member(s), 26 organism(s) | cellular organisms | hemoglobin beta-like protein, partial [Apibacter sp. B3889] | 269 | 269 | 100% | 3e-90 | 87.76% | 162 | WP_221411036.1 |
| 20 member(s), 12 organism(s) | carnivores | hemoglobin subunit beta isoform X3 [Neomonachus schaui… | 267 | 267 | 100% | 1e-89 | 86.39% | 147 | XP_044775197.1 |

## BLAST Result Summary

| Parameter | Value |
|---|---|
| Query Length | 147 aa |
| Query Coverage | 100% |
| Percent Identity | 86% |
| E-value | 1e-97 |
| Alignment Score | 288 bits |
| Top Match | Hb Monza Protein (Homo sapiens) |

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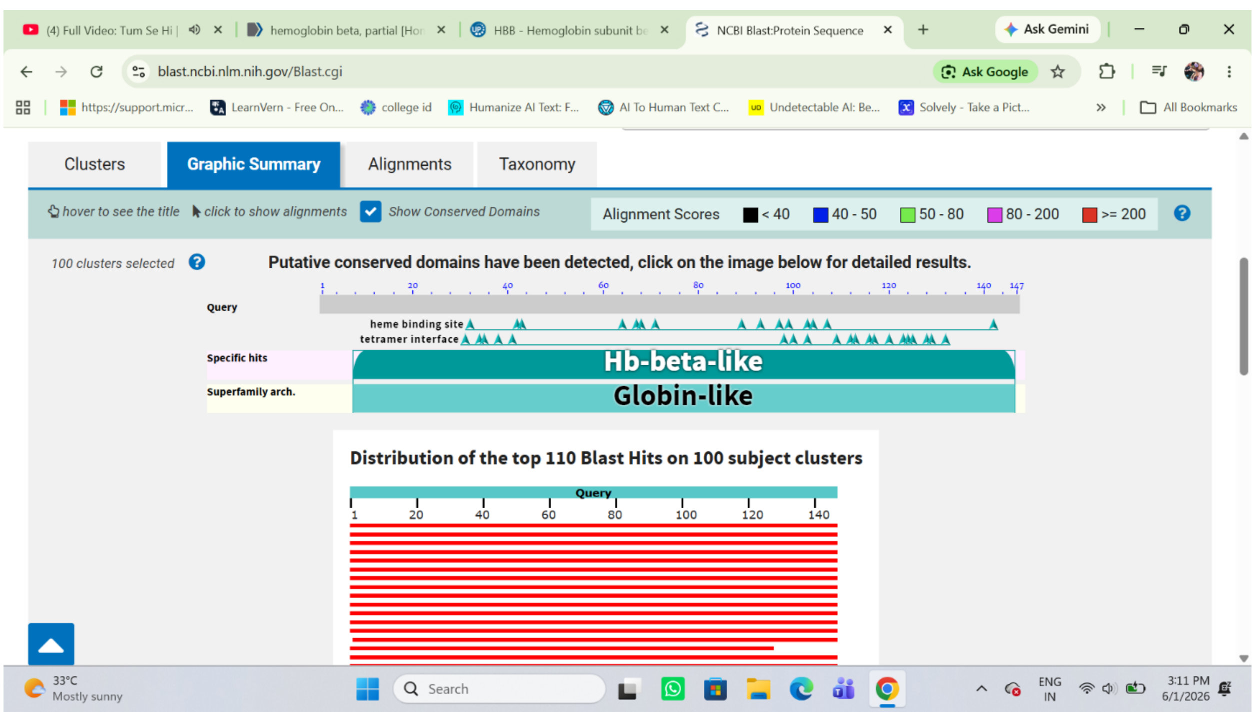
## Graphic Summary

The graphic summary showed significant alignments between the query sequence and homologous protein sequences. Conserved globin domains and heme-binding regions were detected.

## Screenshot 4: Graphic Summary

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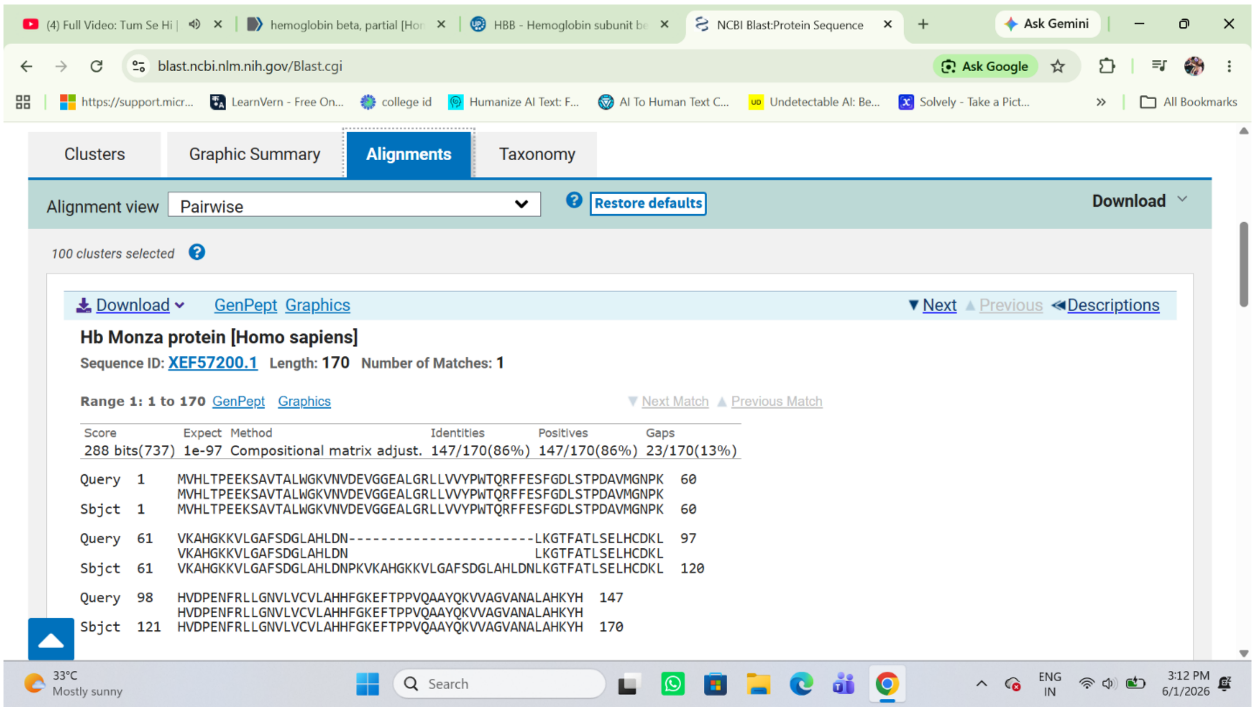
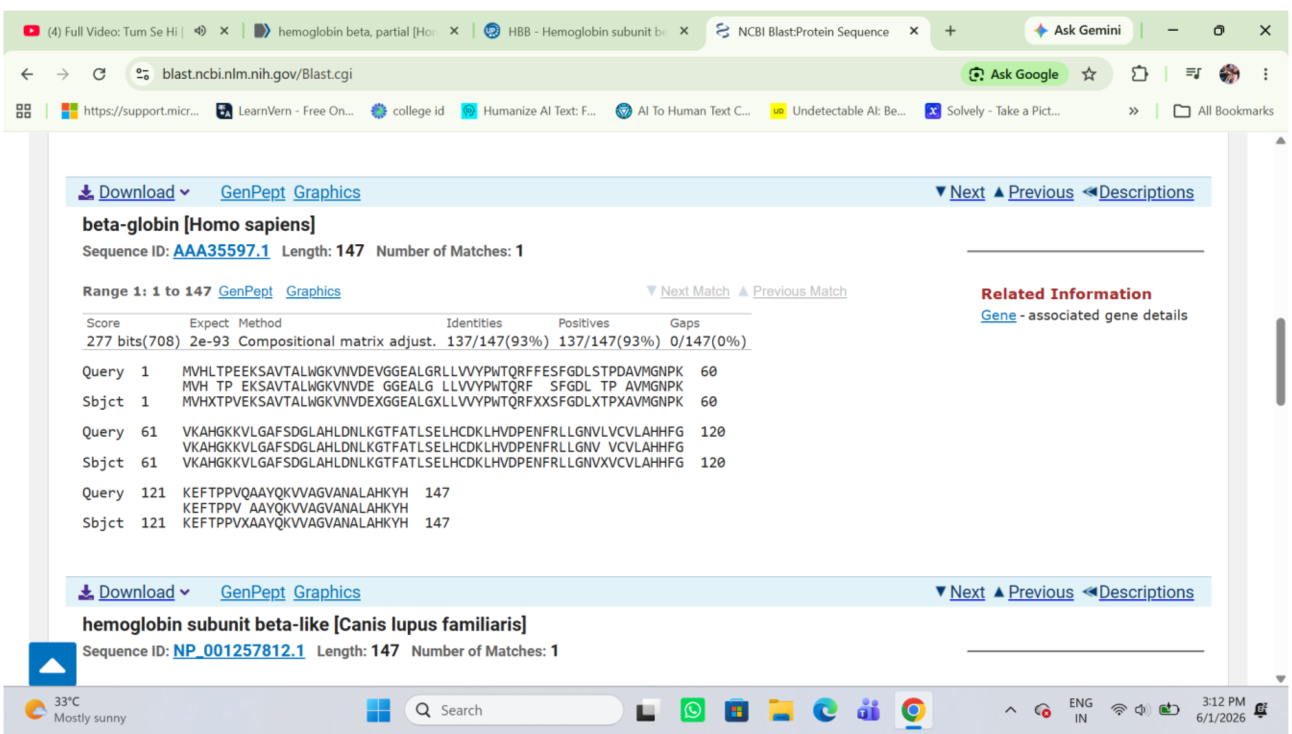
## Alignment Analysis

The pairwise alignment showed a high degree of similarity between the query sequence and the matched protein sequence. The low E-value indicates that the alignment is statistically significant and not due to random chance.

## Screenshot 5: Alignment Results



**Alignment Statistics**

| Parameter | Value |
|---|---|
| Score | 288 bits |
| E-value | 1e-97 |
| Identities | 147/170 (86%) |
| Positives | 147/170 (86%) |
| Gaps | 23/170 (13%) |

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**Discussion**

The BLASTP analysis identified homologous hemoglobin proteins with high sequence similarity. The query coverage of 100% indicates that the complete protein sequence was aligned with the subject sequence. An identity value of 86% suggests strong evolutionary conservation among globin proteins. The extremely low E-value (1e-97) confirms that the observed similarity is highly significant. Conserved globin domains and heme-binding sites indicate the functional importance of the protein.

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**Conclusion**

BLAST analysis successfully identified homologous protein sequences related to Human Hemoglobin Beta Protein. The results demonstrated high sequence similarity, complete query coverage, and significant alignment scores, confirming the conserved nature of hemoglobin proteins among related organisms.

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**References**


1. UniProt Protein Database.
2. NCBI BLAST (Basic Local Alignment Search Tool).
3. National Center for Biotechnology Information (NCBI).